

AWS Developer Certification

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The certification cover over **40 AWS services**
Creation resources in the cloud cost money.

Creating and AWS account require an email for the root user and and AWS account name. You also need a password to authenticate with your root email.

Cloud computing is the on-demand delivery of compute power, database storage, applications, and other IT resources.
Through a cloud services platform with pay-as-you-go pricing
You can provision exactly the right type and size of computing resources you need

AWS is one of the many cloud provider available on the market!

The five characteristics of cloud computing are the following:

- **On-demand services:** without the need of human interaction
- **Broad network access:** resources can be accessed everywhere
- **Multi-tenancy and resource pooling** multiple customer are served on the same resources
- **Rapid elasticity and scalability** easily create and dispose of resources
- **Measured service** user pay for what they use

AWS has 3 pricing fundamentals:

- **Compute:** Pay for your computational time
- **Storage:** Pay for data stored
- **Data transferred OUT of the cloud**

Shared responsibility Model

You are responsible for the security **IN** the cloud
AWS is responsible for the security **OF** the cloud

1 IAM

AWS IAM is one of the few **global** services

On AWS you apply the **least privilege** principle for the user and services IAM policy

1.1 IAM Security Tools

IAM Credentials Report (account-level) is a service used to lists of all your account's users and the status of their various credentials.

IAM Access Advisor (user-level) is a service used to shows the service permissions granted to a user and when those services were last accessed. You can use this information to revise your policies.

Don't use the root account except for AWS account setup

EC2 User Data script run with **root user**

2 EC2

AWS has the following naming convention:

m5.2xlarge

- m: instance class
- 5: generation (AWS improves them over time)
- 2xlarge: size within the instance class

2.1 Security Groups

Security Groups control how traffic is allowed into or out of our instances.

Security groups only contain **allow** rules.

Security groups rules can reference by IP or by security group.

They controll both inbound and outbound traffic Security Group are linked to the specific **region and VPC** where they are created

If your application is not accessible (time out), then it's a security group issue.

Usage beyond Savings Plans is billed at the On-Demand price.

2.2 EBS Volume

They can only be mounted to one instance at a time.

They are bound to a specific availability zone.

It's locked to an Availability Zone, to move a volume you need to take a snapshot.

2.3 EBS Snapshots

Make a backup (snapshot) of your EBS volume at a point in time.

You can move your EBS snapshot to a cheaper archive tier.

You can specify a retention policy (1 day - 1 year) for your deleted EBS snapshot

2.4 AMI

AMI are customization of an EC2 instance **EC2 Image Builder** is the service use to schedule and organize AMI creation

2.5 EC2 Image Builder

Used to automate the creation of Virtual Machines or container images.

Can be run on schedule

2.6 EC2 Instance store

high performance disk attached to EC2 instances

2.7 EFS

EFS can be mounthed to multiple target (100s of EC2 at the same time) and is not available zone bounded

2.8 AWS FSx

Launch 3th party high performance sys on AWS.
FSx is a windows native shared file system

3 AWS S3

Maximum size of an s3 object is 50TB.
For uploadin object bigger than 5Gb we need to use **multi-part upload**.

S3 bucket policy are used for cross-account bucket/object sharing.

S3 bucket versioning is enabled at bucket level

3.1 IAM Access Analyzer for S3

Ensure that only the correct people have access to your S3 bucket

3.2 AWS Snowball

On-edge device to **collect and process** data and **migrate** them on AWS

3.3 AWS Storage Gateway

Bridge between the on premise data and AWS data storage option (S3, EBS,...)

4 Database solutions

4.1 AWS RDS

RDS is a **Relational** Database that use SQL data formatting.
Amazon Aurora RDS engine have the **serverless** option, where we don't provision a capacity for the daatabase but the size is automatically managed by AWS

4.2 ElastiCache

ElastiCache is the managed cache solution for AWS

4.3 DynamoDB

The noSQL database service for AWS with serverless database capability. **DAX** is the in-memory cache service for dynamoDB.

4.4 Global Tables

Make a DyamoDB database accessible with low latency on multiple region

4.5 Redshift

OLAP (online analytical process) database with serverless capability

4.6 Amazon EMR

Elastic MapReduce DB service that manage hundred of EC2 instances with the possibility to fire "spot instances"

4.7 Amazon QuickSight

Serverless machine learning-powered service use to create interactive dashboard

4.8 DocumentDB

MongoDB inspired service used to store document

4.9 Amazon Neptune

Fully managed **graph** database

4.10 Amazon Timestream

Fully managed serverless **time series** database

4.11 Amazon Lightsail

Virtual servers, storage, databases, and networking.
Good for people with little cloud experience

5 Infrastructure at scale